

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
Student Name:	M.S.Dinesh	Reg. No.:	192424016
Section / Batch:	I	Date:	28.07.2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT**Code of Conduct**

I (M.S.Dinesh / 192424016) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.S.Dinesh

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q33. Library System – Linear Search

Problem Statement

A library system stores **book IDs in an unsorted list** because books are frequently added and removed. Users search for a specific book using a basic search feature.

a) Explain how Linear Search is applied in this scenario.

Since the book IDs are stored in an **unsorted list**, the system cannot use Binary Search. Instead, it applies **Linear Search**, which checks each book ID one by one from the beginning until the required book ID is found or the list ends.

Steps of Linear Search

1. Start from the first book ID.
2. Compare it with the target book ID.
3. If they match, return the position of the book.
4. Otherwise, move to the next book ID.
5. Repeat until the book is found or all IDs have been checked.

Example

Book IDs:

[105, 212, 187, 340, 156, 421]

Target Book ID:

340

Search Process:

Step	Book ID	Comparison	Result
1	105	$105 = 340?$	No
2	212	$212 = 340?$	No
3	187	$187 = 340?$	No
4	340	$340 = 340?$	Book Found

The book is found after **4 comparisons**.

b) Analyze the Best, Average, and Worst-Case Complexities

Case	Description	Comparisons	Time Complexity
Best Case	Book is at the first position	1	O(1)
Average Case	Book is somewhere in the middle	$n/2$	O(n)
Worst Case	Book is at the last position or not present	n	O(n)

Best Case – O(1)

The required book is the first element.

Example:

[340, 105, 212, 187, 156]

Only **1 comparison** is needed.

Average Case – $O(n)$

The required book is located near the middle.

Approximately:

[
 $\frac{n}{2}$
]

comparisons are performed.

Worst Case – $O(n)$

The required book is at the end of the list or does not exist.

Example:

[105, 212, 187, 156, 421, 340]

or

Searching for 500 (not available)

All elements must be checked.

Space Complexity

Linear Search uses only a few extra variables.

Auxiliary Space = $O(1)$

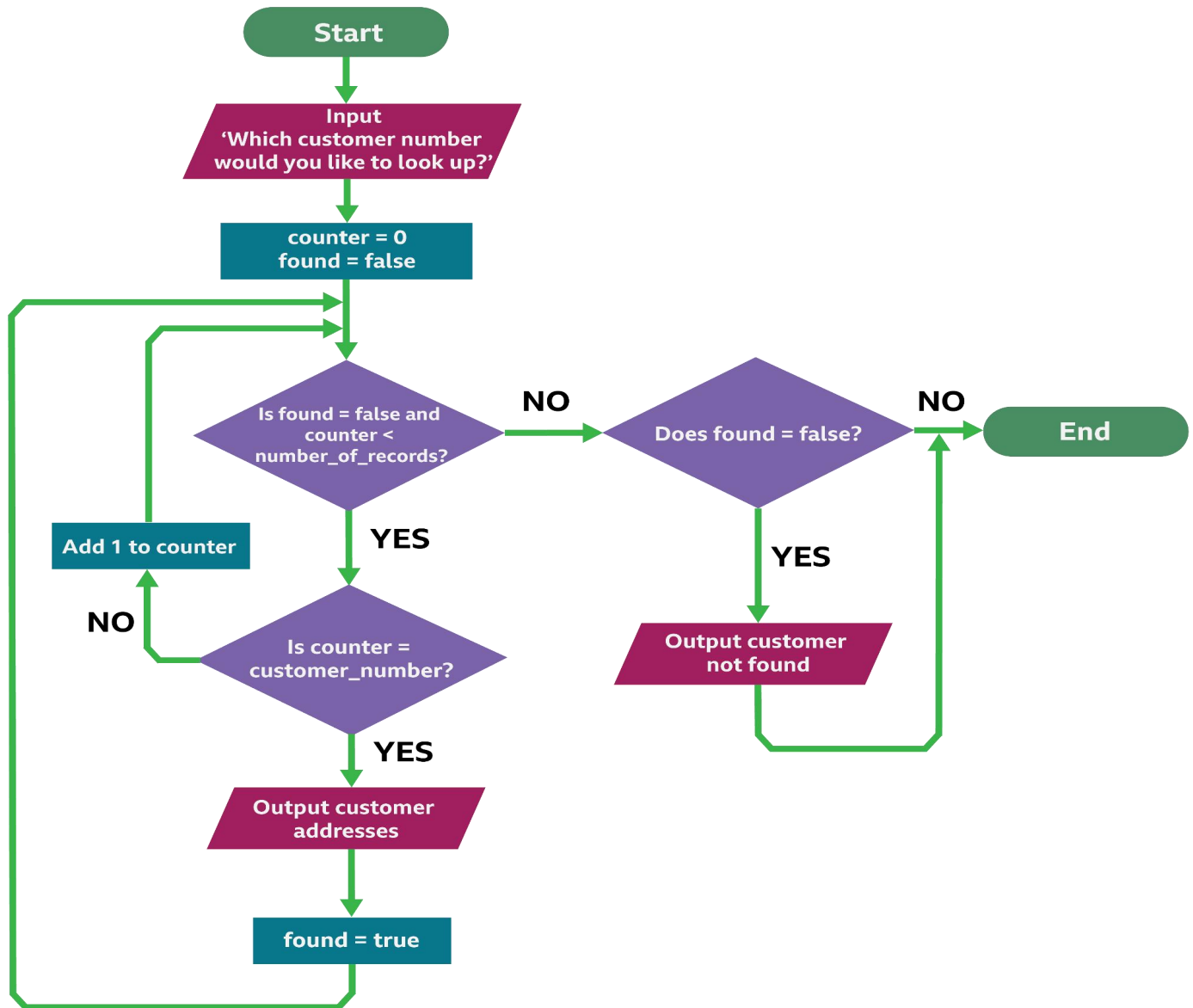


Figure 1 :Linear Search Flowchart

c) Justify whether sorting would improve system efficiency.

Yes, sorting can improve search efficiency, but it depends on the situation.

Advantages of Sorting

- Enables the use of **Binary Search**.
- Binary Search reduces search time from $O(n)$ to $O(\log n)$.
- Faster searching for large datasets.
- Better performance when searches are frequent.
- Efficient retrieval of book records.

Disadvantages of Sorting

- Frequent additions and deletions require the list to be sorted again.
- Maintaining sorted order increases update time.
- Sorting itself takes $O(n \log n)$ time.
- Extra processing is required after every insertion or deletion.
- Not efficient if the library database changes very often.

Justification

In this library system, books are **frequently added and removed**, making the list highly dynamic. Keeping the list sorted after every update would introduce additional overhead. Therefore, **Linear Search is a practical choice for frequently changing data**. However, if the list becomes mostly stable and users perform many searches, sorting the list and using **Binary Search** would significantly improve search efficiency.

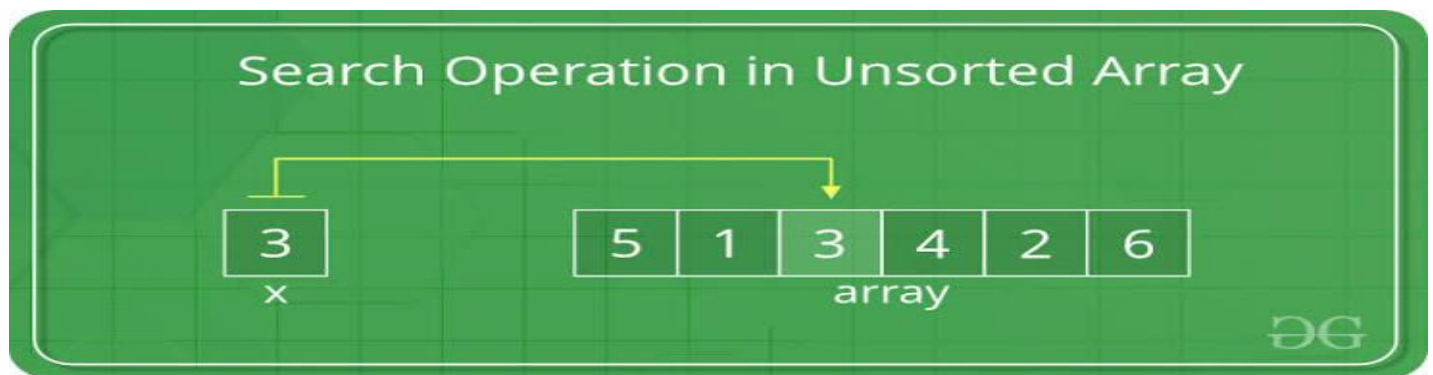


Figure 2 : Unsorted Array Linear Search

Final Answer

- a) Linear Search checks each book ID sequentially until the required book is found or

the list ends.

- **b) Time Complexity:**
 - **Best Case: $O(1)$**
 - **Average Case: $O(n)$**
 - **Worst Case: $O(n)$**
 - **Space Complexity: $O(1)$**
- **c) Sorting improves search efficiency by allowing **Binary Search ($O(\log n)$)**, but because the library data changes frequently, maintaining a sorted list is costly. Hence, **Linear Search is more suitable for this scenario**, while sorting is beneficial only when searches are frequent and updates are infrequent.**